

# Agentic Analytics in Production

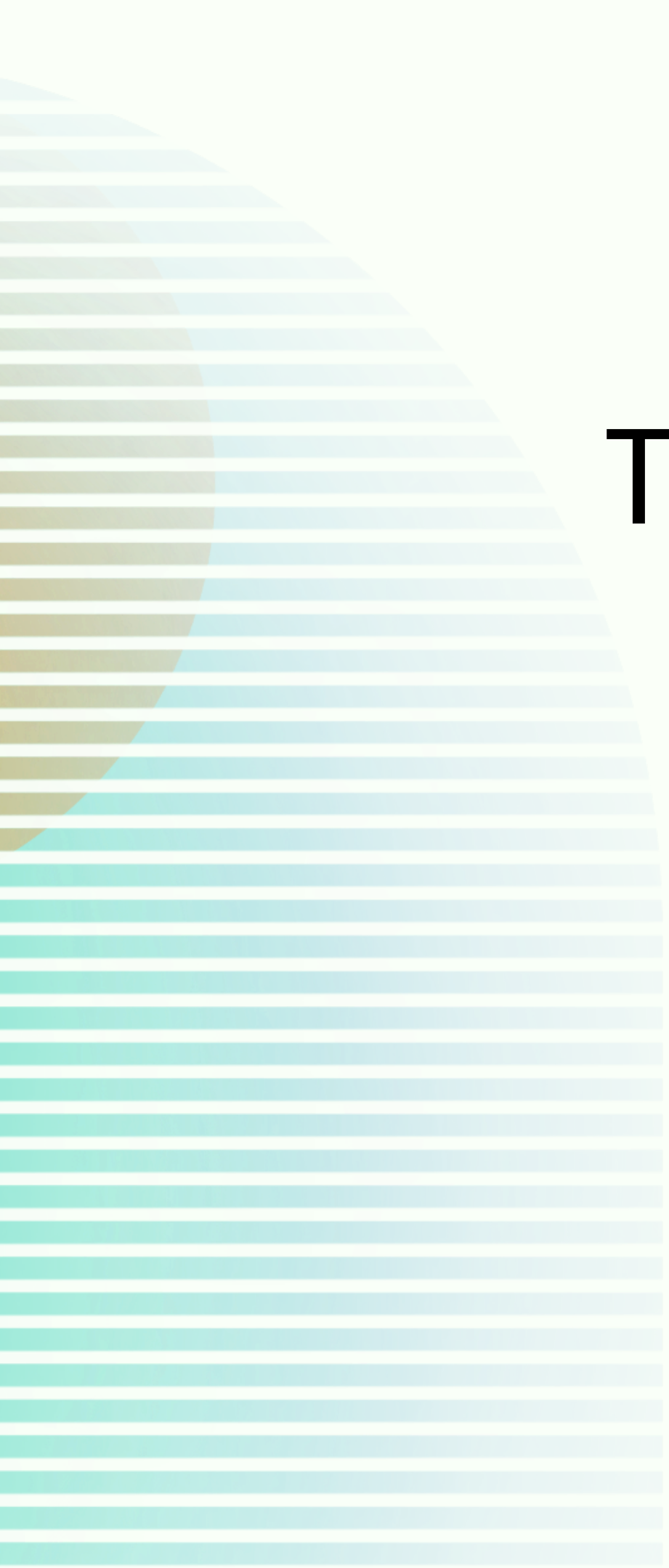
## Lessons From Building a Talk-to-Your-Data Slackbot

BY ANDRES VOURAKIS

# While We Wait

*Let me know in the chat*

1. Where are you calling from?
2. How are you using AI in your day-to-day?



# The Agenda for Today

- Quick introduction
- Gaining an intuition for Agentic Analytics
- Breaking down the architecture of the talk-to-your-data Slackbot
- What it takes to go from prototype to production
- How data scientists can stay ahead
- Q&A (15-20 mins)



# ABOUT ME



ANDRES VOURAKIS

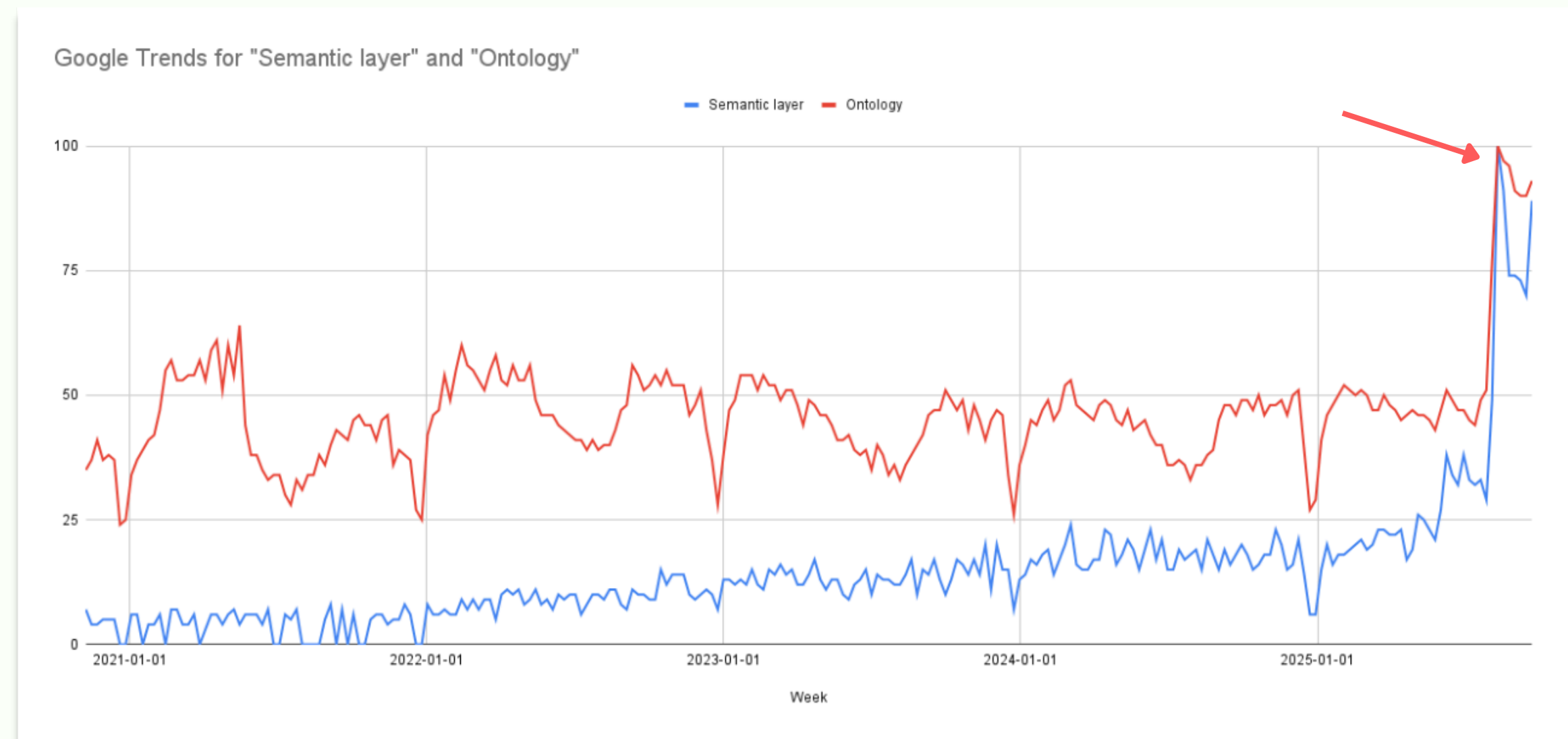
**Current** Senior Data Scientist @ Nextory,  
Founder of “Future Proof DS”

**Past** Data Scientist @ Epidemic Sound,  
Business Analyst, Data Analyst  
B.S. Computer Science, specialised in ML





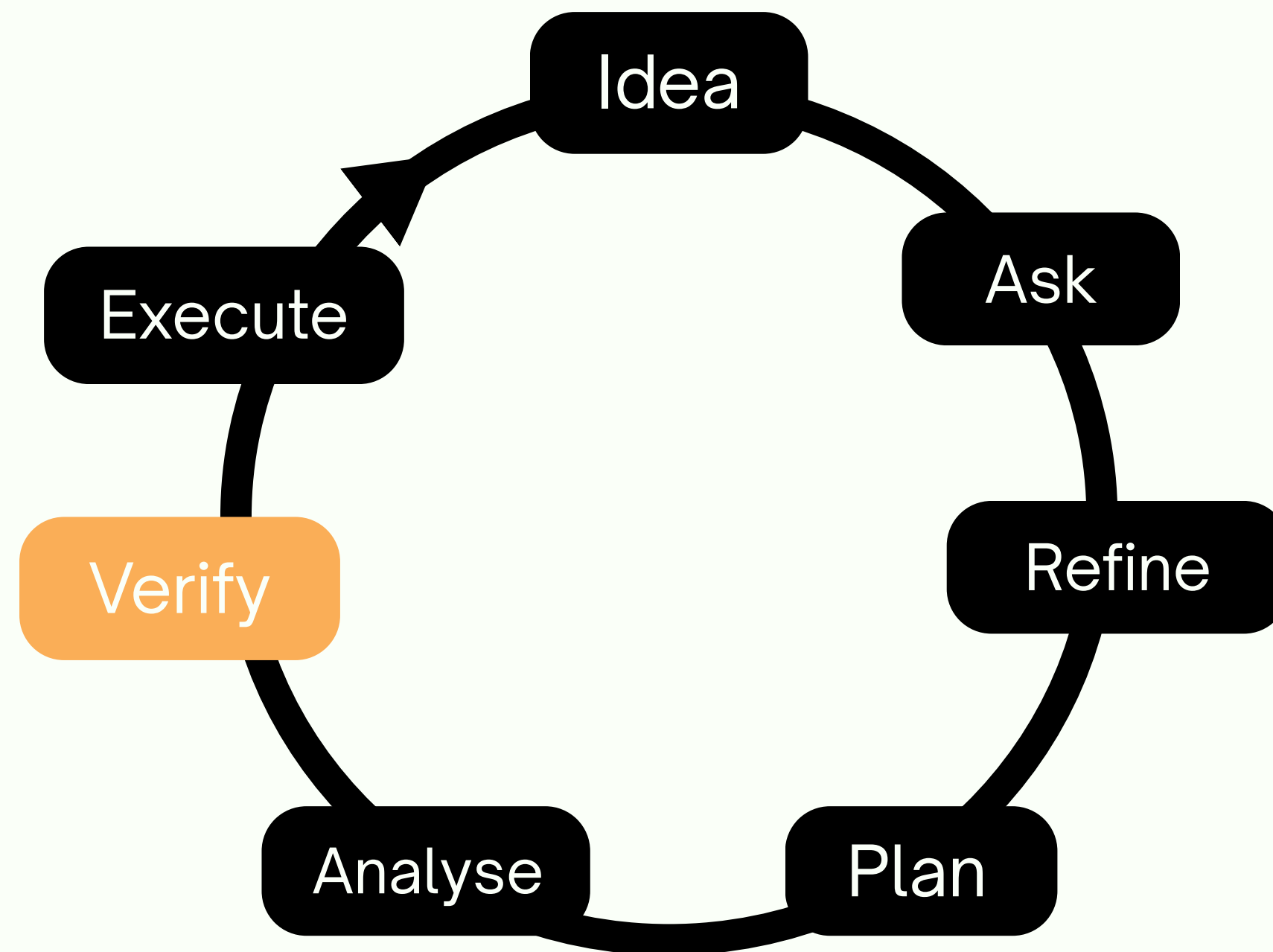
# More than just about Agents...



# What *exactly* is Agentic Analytics?



## TYPICAL ANALYTICS WORKFLOW

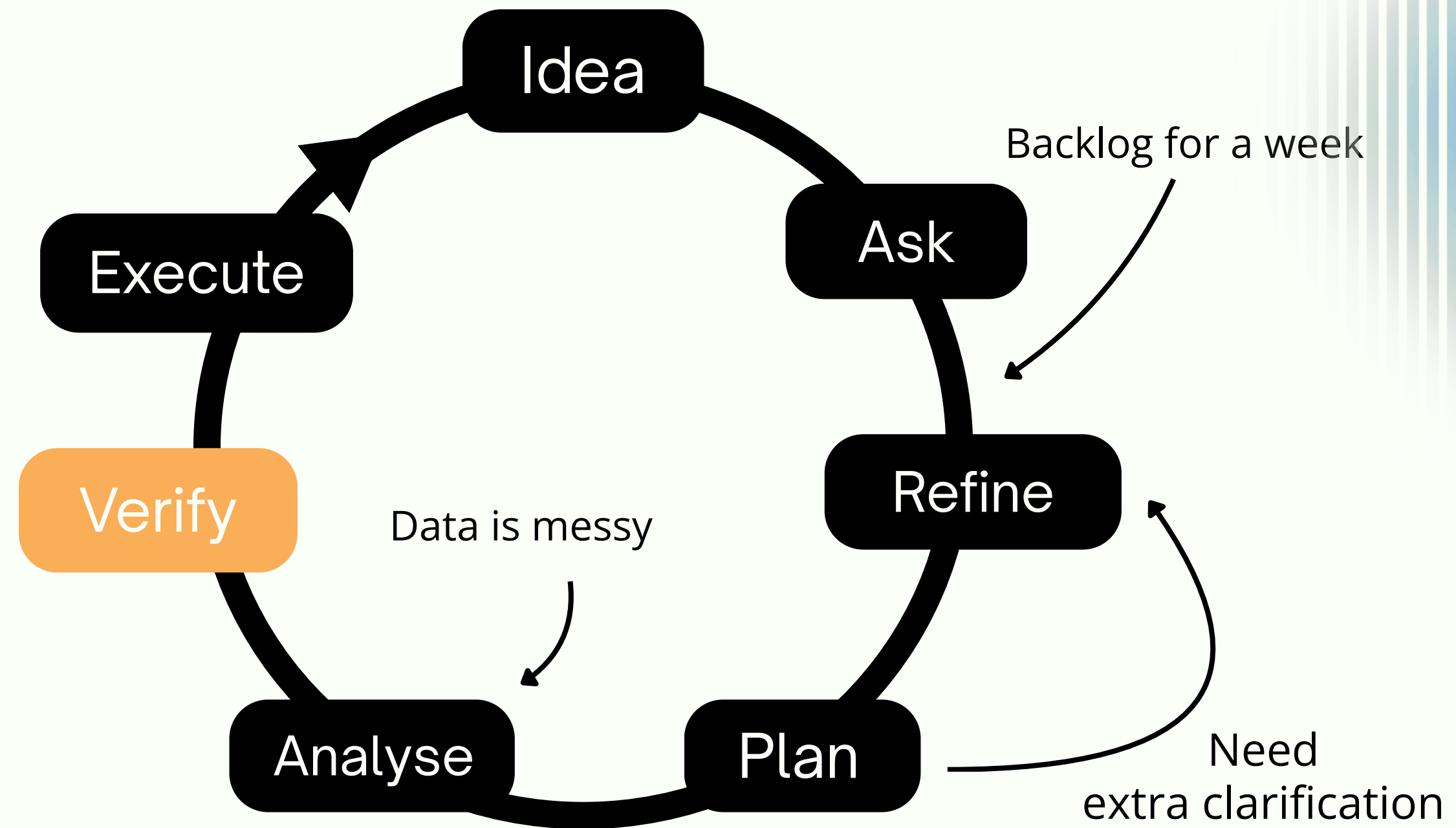




# THE DECISION-LATENCY PROBLEM

It can take a long time from the initial idea to verifying it with insights.

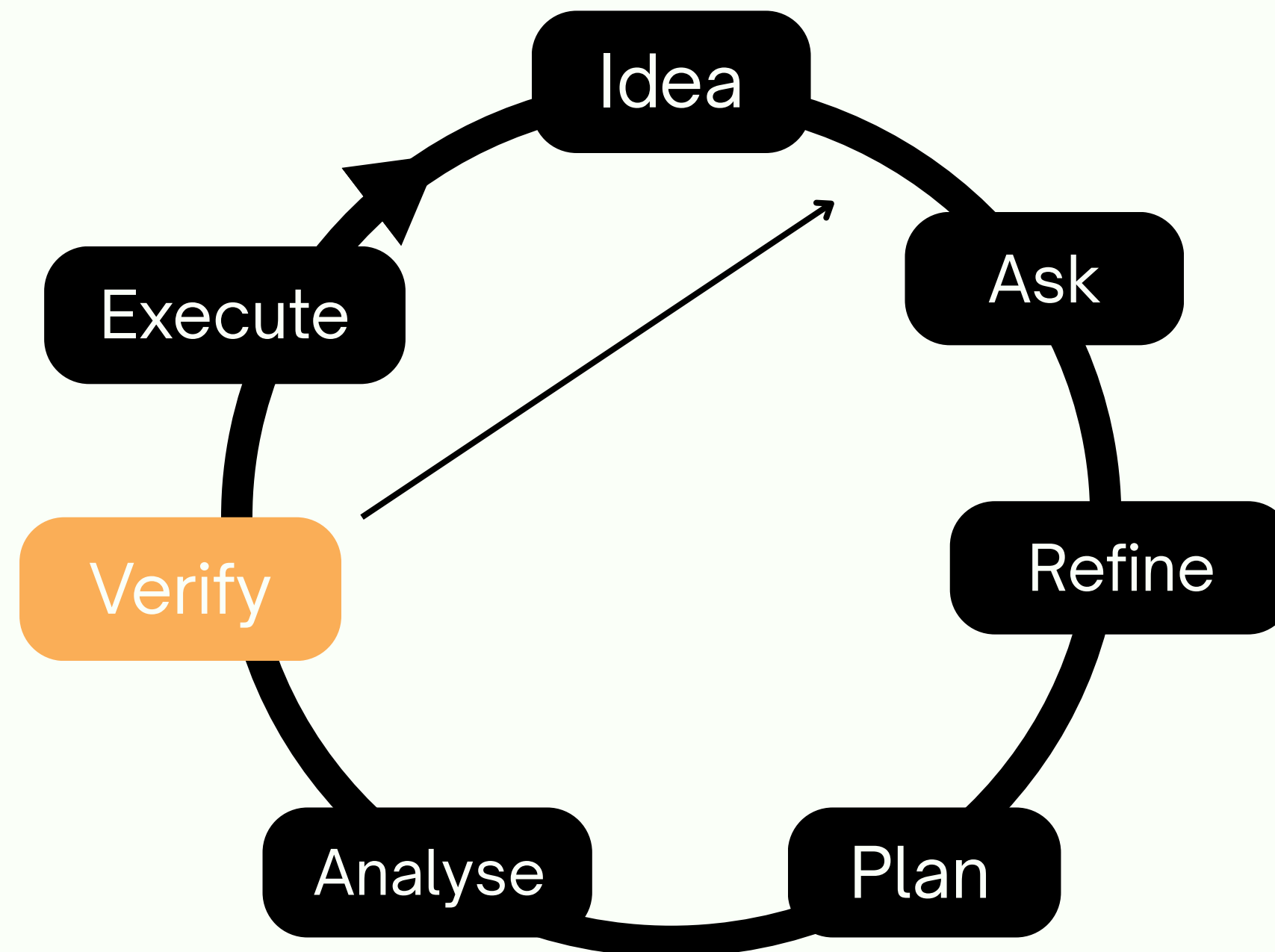
## TYPICAL ANALYTICS WORKFLOW



# THE DECISION-LATENCY PROBLEM

Agentic analytics is an enabler for early insights.

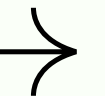
## TYPICAL ANALYTICS WORKFLOW



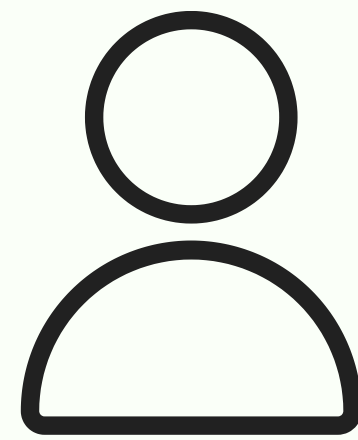
# What *exactly* is Agentic Analytics?



<https://www.leewayhertz.com/ai-agents-for-data-analysis>

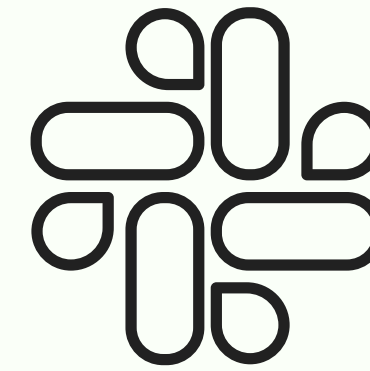
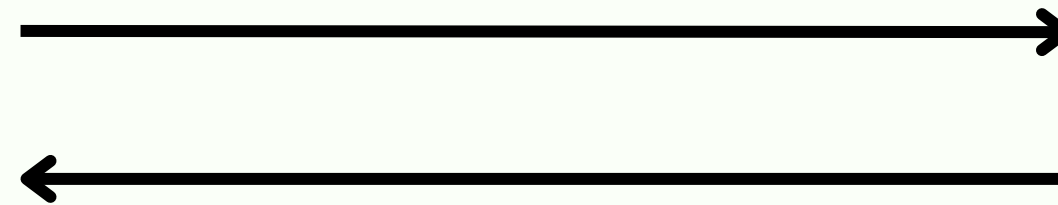


# Talk-to-your-data Slackbot

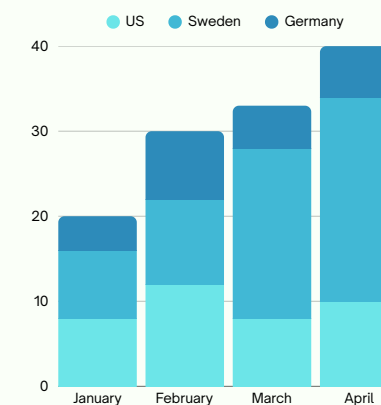


Non-technical  
Stakeholder

**Prompt:** Give me a chart for our monthly revenue  
split by our top 3 markets

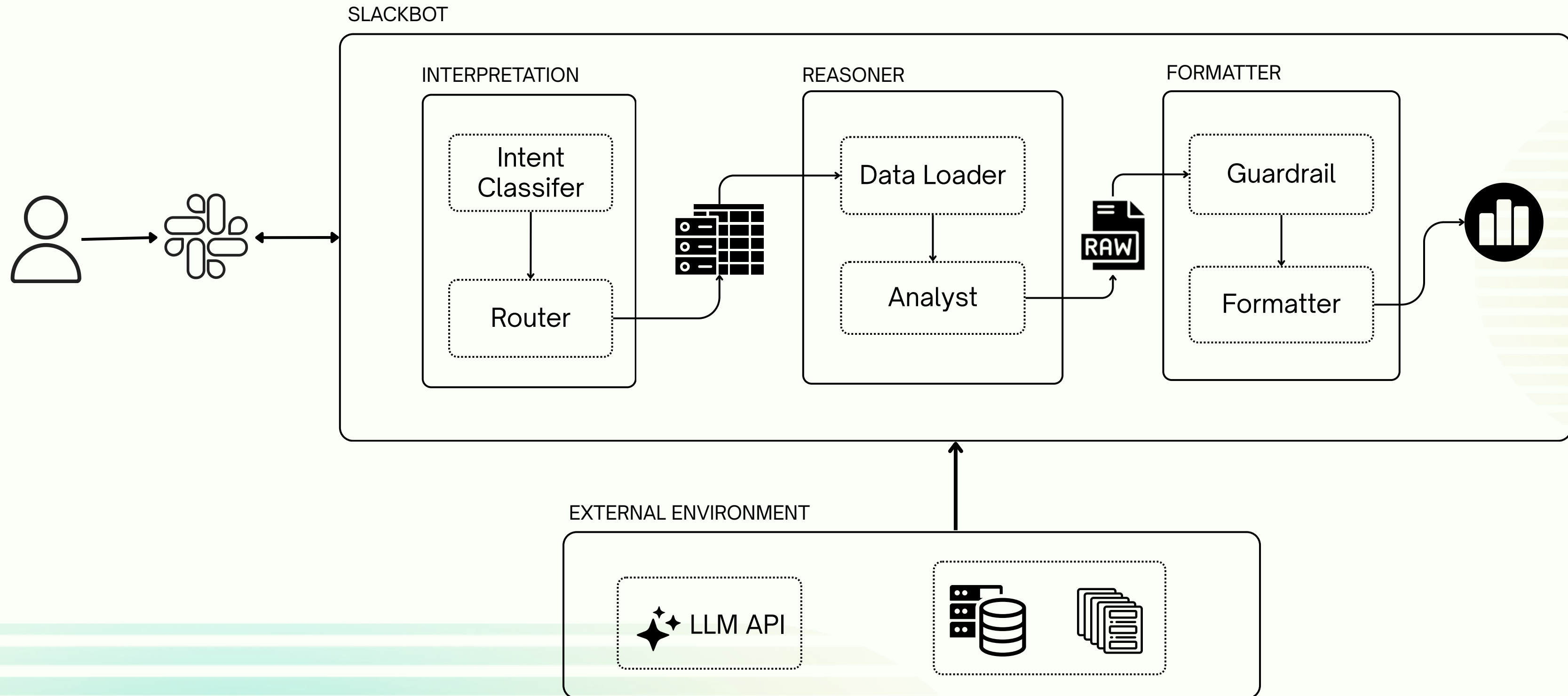


Slack

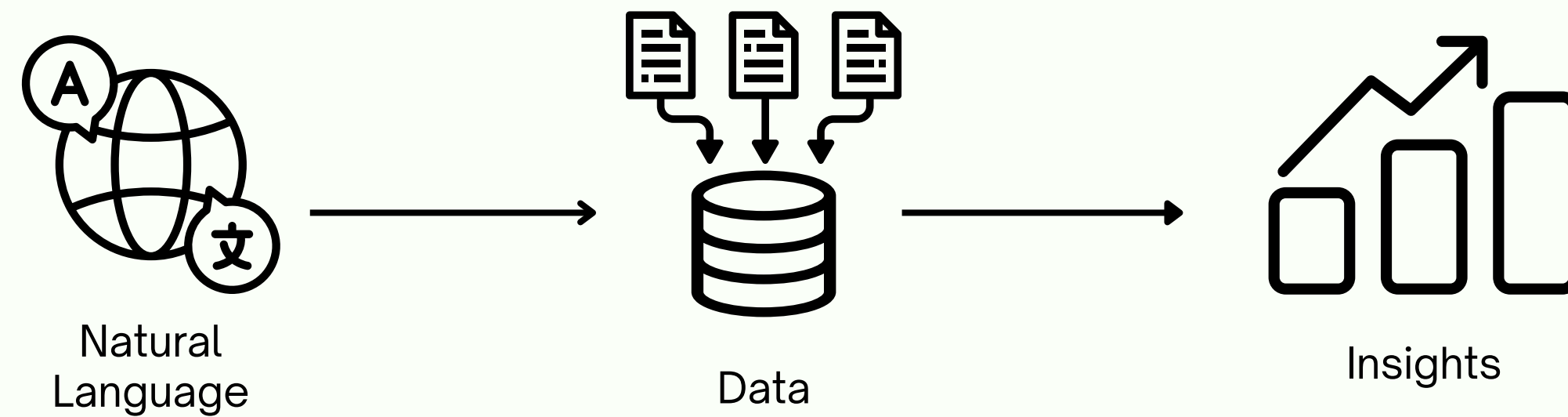




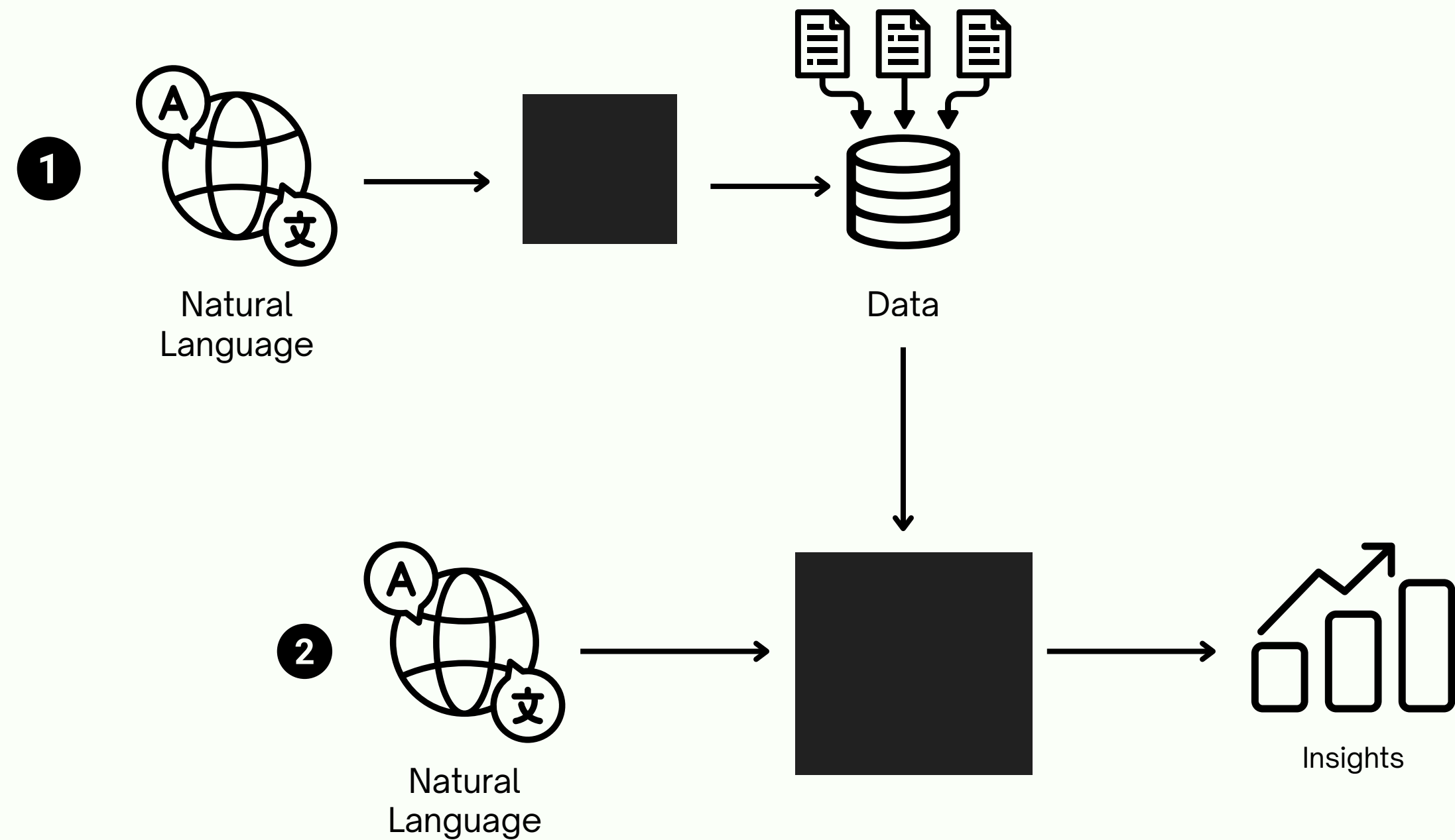
# Our minimum viable product (MVP)



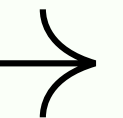
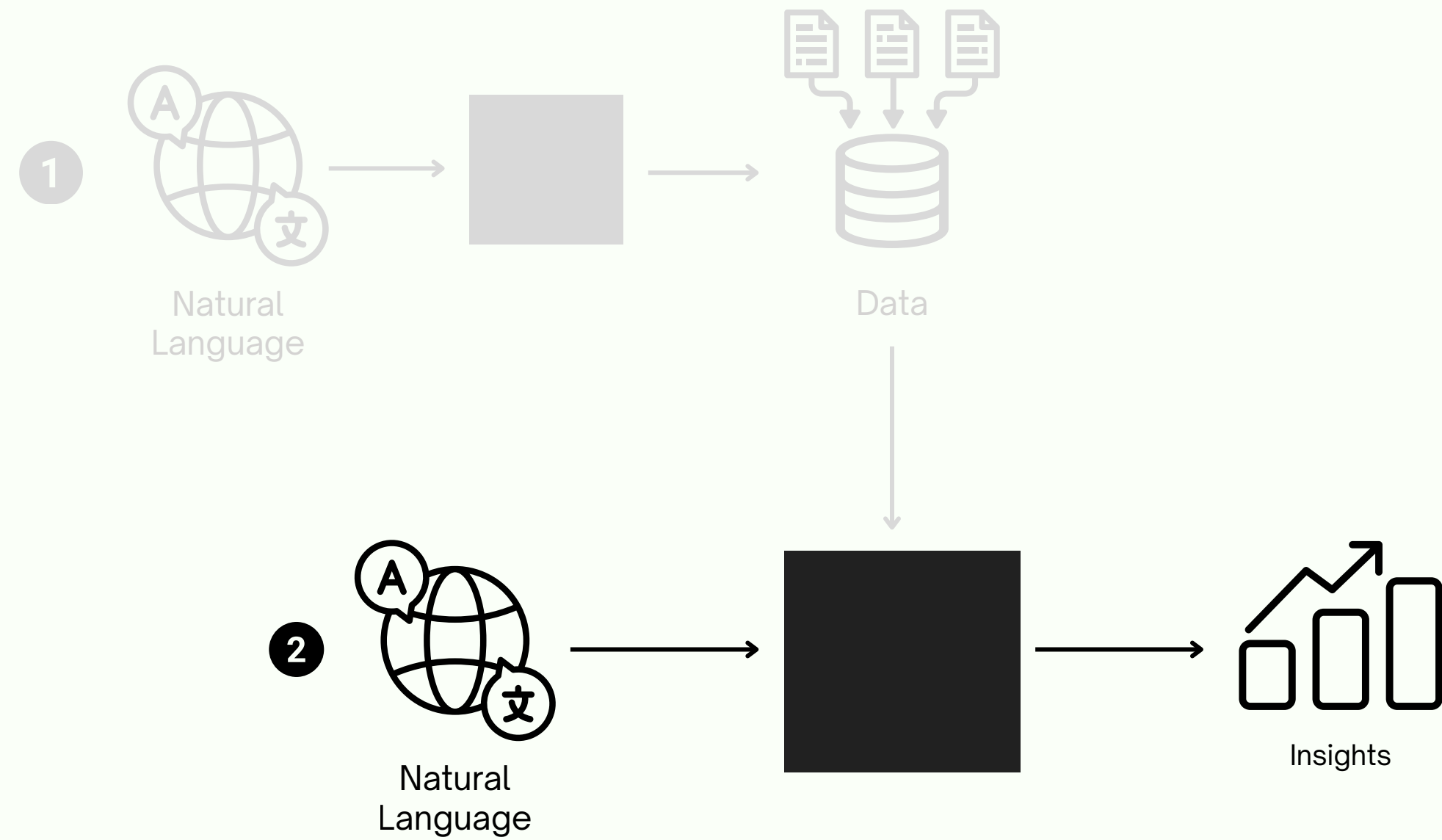
# The simplified flow...



# Breaking it down into subsystems...

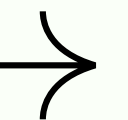
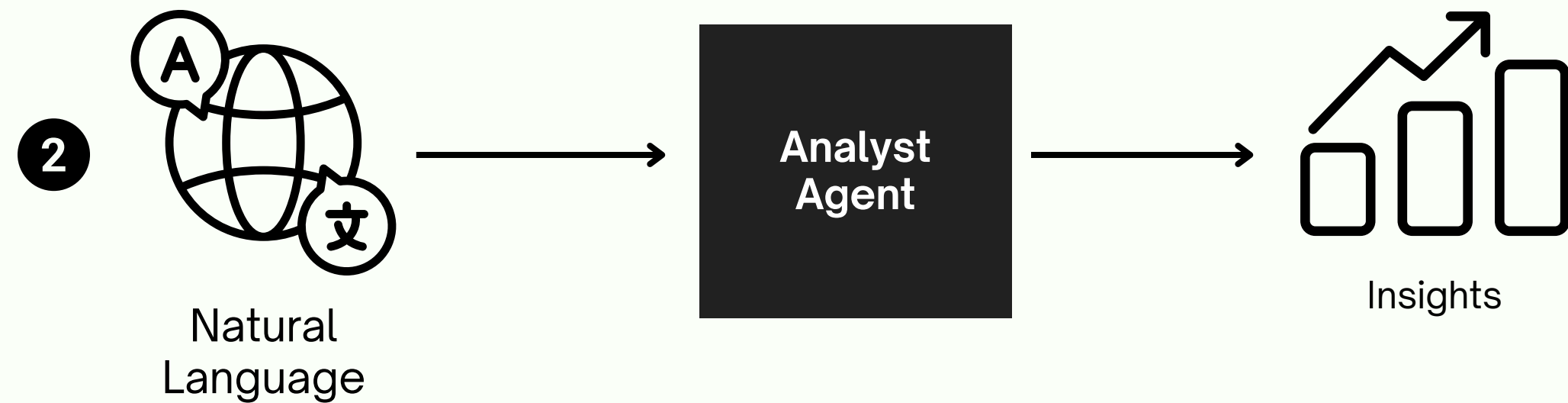


# SUBSYSTEM #2



## SOLVING: ANALYST AGENT

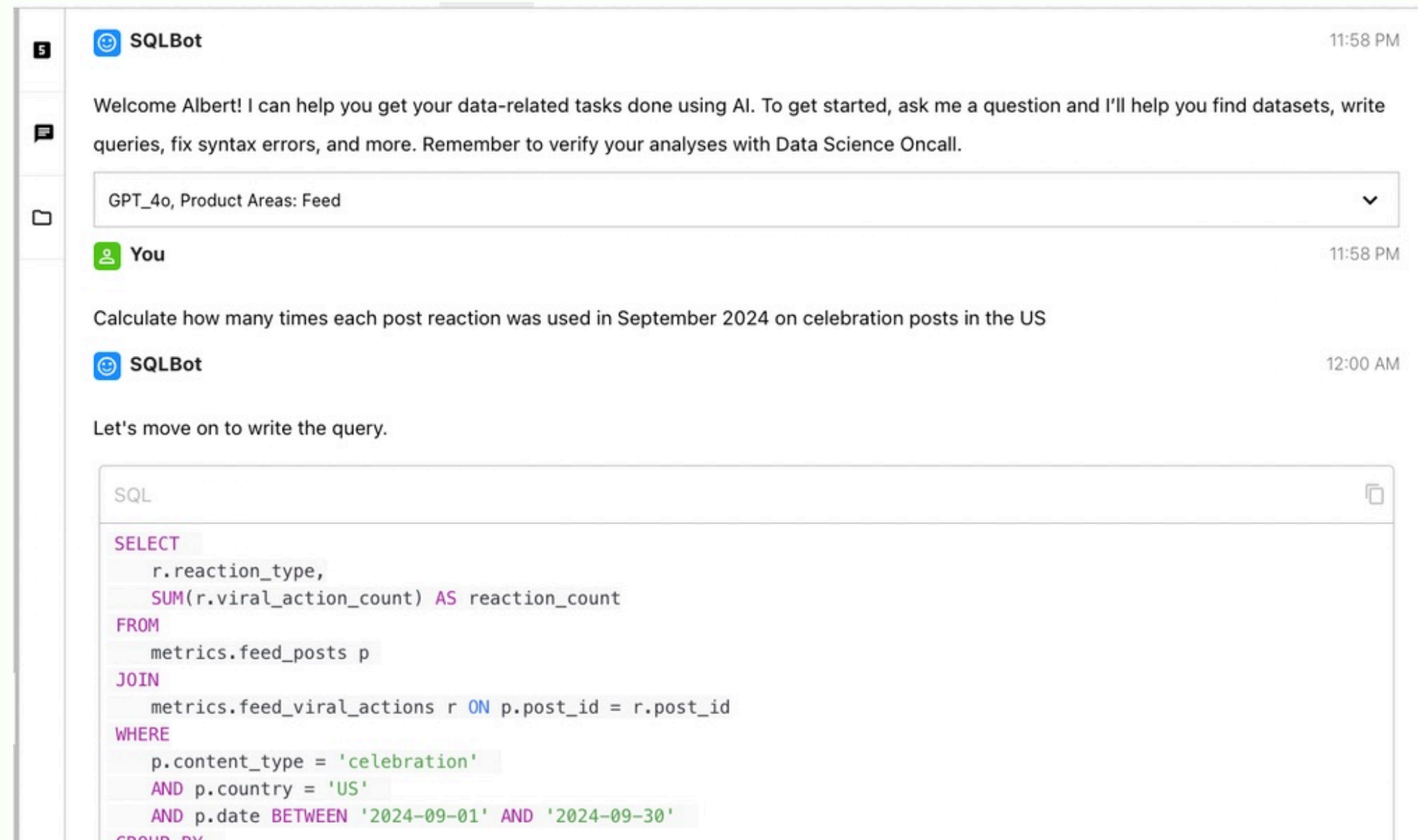
**Objective:** Transforming and analysing the data based on the user's question.





## SOLVING: ANALYST AGENT

# The obvious first idea: Text-to-SQL...



# SOLVING: ANALYST AGENT

## The challenges with this approach for an MVP:

- It only covers query generation, not the full insight workflow
- It requires solving the hardest problem first (joins, schema ambiguity)
- Needs solid failure and retry handling from day one

5

SQLBot

11:58 PM

Welcome Albert! I can help you get your data-related tasks done using AI. To get started, ask me a question and I'll help you find datasets, write queries, fix syntax errors, and more. Remember to verify your analyses with Data Science Oncall.

GPT\_4o, Product Areas: Feed

You

11:58 PM

Calculate how many times each post reaction was used in September 2024 on celebration posts in the US

SQLBot

12:00 AM

Let's move on to write the query.

SQL

```
SELECT
  r.reaction_type,
  SUM(r.viral_action_count) AS reaction_count
FROM
  metrics.feed_posts p
JOIN
  metrics.feed_viral_actions r ON p.post_id = r.post_id
WHERE
  p.content_type = 'celebration'
  AND p.country = 'US'
  AND p.date BETWEEN '2024-09-01' AND '2024-09-30'
GROUP BY
  r.reaction_type
ORDER BY
  reaction_count DESC;
```

Verifications

✓ All tables are valid

✓ All columns are valid

✓ No syntax issues

Query Explanation:

The query retrieves the count of each post reaction by joining the 'metrics.feed\_viral\_actions' table with the 'metrics.feed\_posts' table on the 'post\_id'. It filters for September 2024 and US celebration posts to calculate the number of times each post reaction was used.

Here are the tables used in the query:

1. metrics.feed\_posts [\[View on DataHub\]](#)

This dataset hosts metrics related to content creation on LinkedIn, including original posts, reshare posts, group posts, messaging shares, etc.

2. metrics.feed\_viral\_actions [\[View on DataHub\]](#)

This dataset contains metrics related to social gestures and viral actions (such as react, like, comment, etc.) that are taken on feed updates, with the information on both actors and the feed updates which the actions are taken upon.

# SOLVING: ANALYST AGENT

## What PandasAI solves from the start:

- It introduces Python's flexibility
- Built-in support for summaries and charts.
- It's an open-source library



# PandasAI

Release **v3.0.0** ci-core **passing** cd **passing** codecov **91%** [Discord](#) downloads **5M** License **MIT** [Open in Colab](#)

PandasAI is a Python library that makes it easy to ask questions to your data in natural language. It helps non-technical users to interact with their data in a more natural way, and it helps technical users to save time, and effort when working with data.

### Getting started

You can find the full documentation for PandasAI [here](#).

### Using the library

#### Python Requirements

Python version **3.8+ <=3.11**

#### Installation

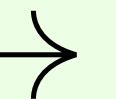
You can install the PandasAI library using pip or poetry.

With pip:



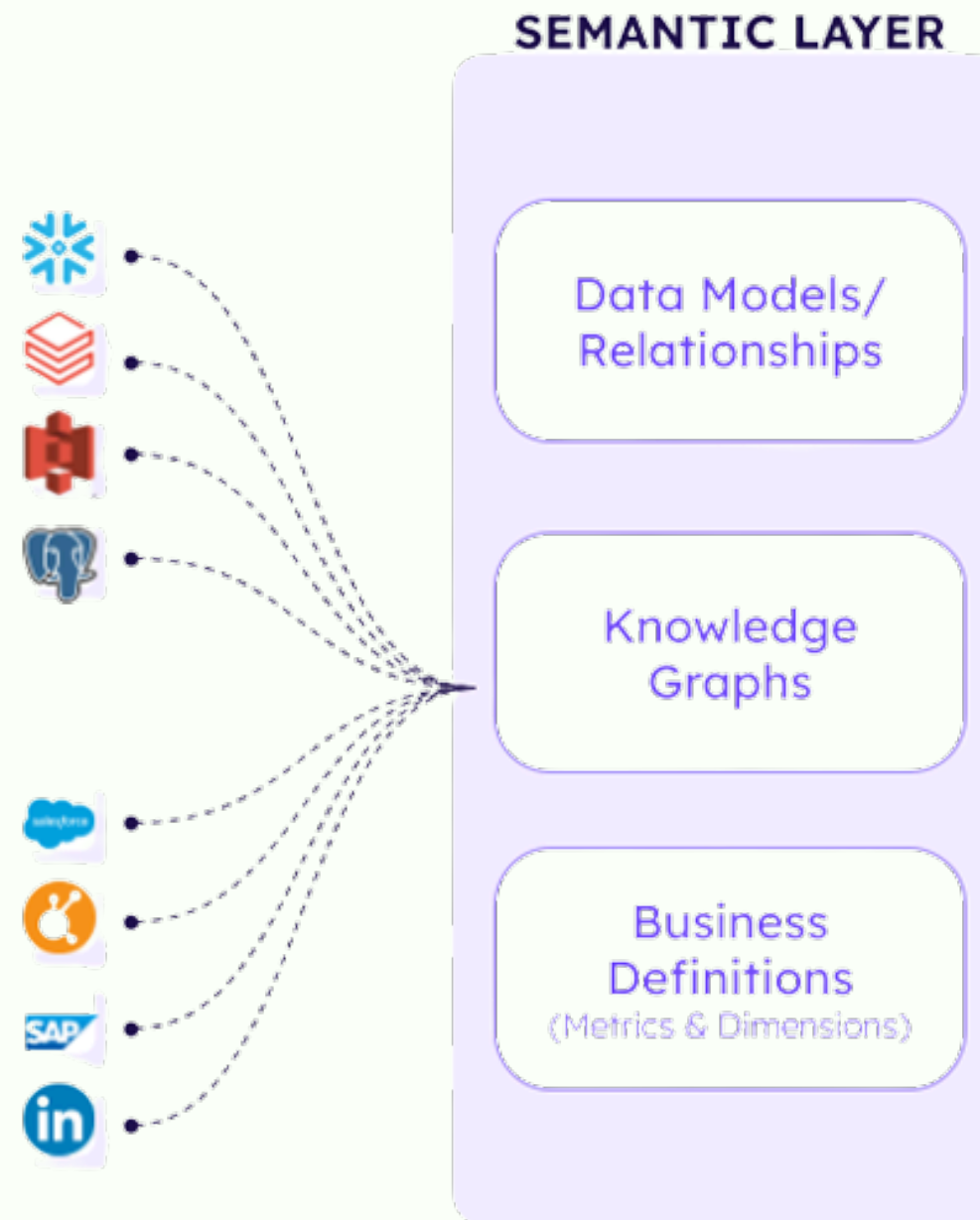
What is the most common *underlying* reason agentic analytics systems fail to delivery useful insights?

- A** Struggle to generate the perfect code or query
- B** Use a data source that isn't actually relevant
- C** Misunderstand the business context behind the question
- D** Present the answer in a format the user didn't expect

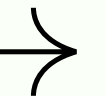




## SIDE NOTE: THE SEMANTIC LAYER



<https://www.tellius.com/resources/blog/is-a-semantic-layer-necessary-for-enterprise-grade-ai-agents>





# SIDE NOTE: THE SEMANTIC LAYER

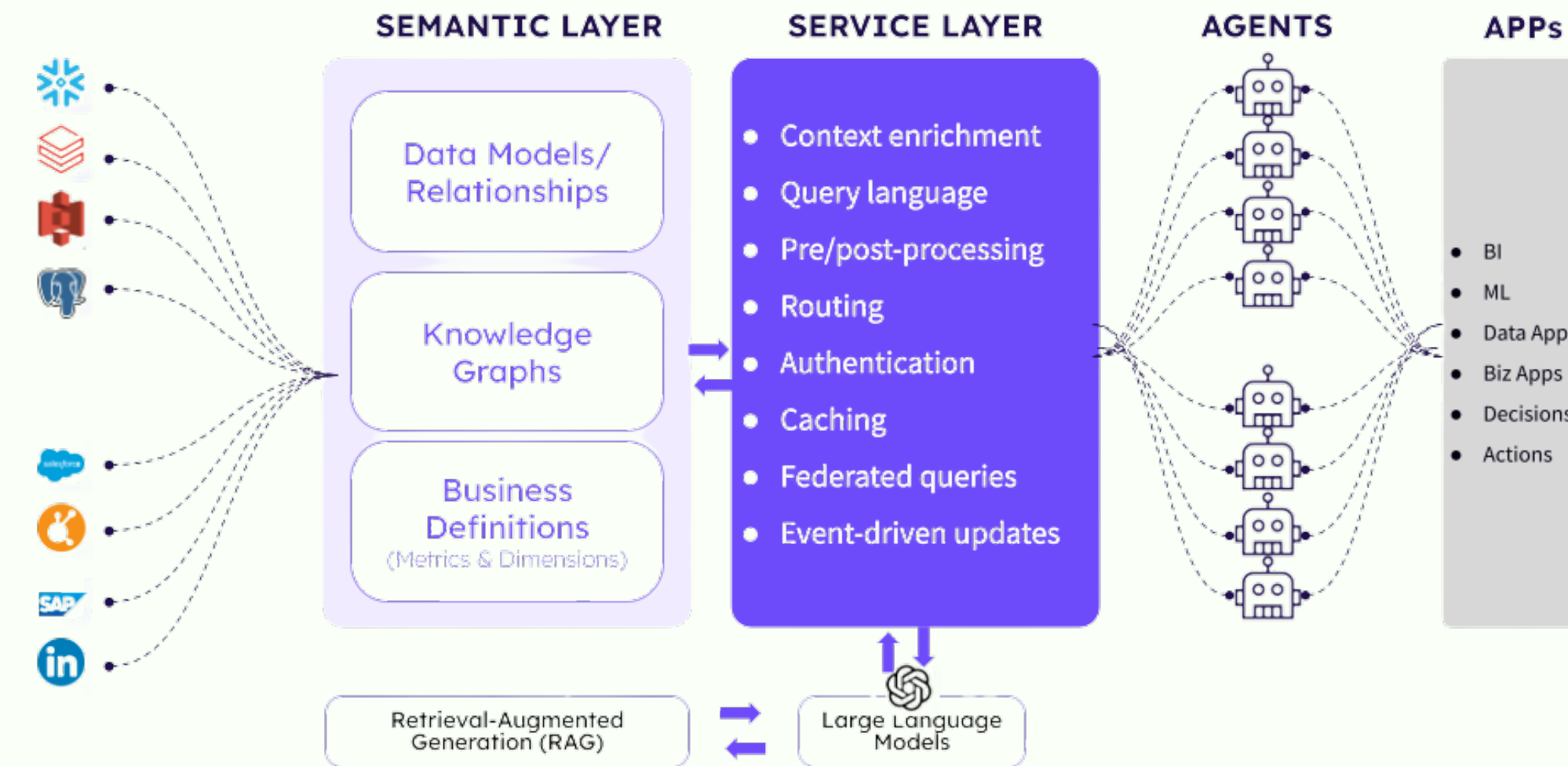
## What goes into a YAML file?

- Metric definitions
- Dimensions and grain
- Accepted filters or breakdowns
- Source table information
- Business-friendly descriptions

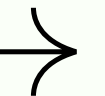
```
revenue_insights.yaml
1 version: 1.0
2 model: revenue_insights
3 description: Defines business metrics for understanding user growth and monetization.
4
5 entities:
6   - name: user
7     primary_key: user_id
8     description: Represents a single customer in the system.
9
10  - name: subscription
11    primary_key: subscription_id
12    description: Represents a paid or trial subscription linked to a user.
13
14  relationships:
15    - from: user
16      to: subscription
17      key: user_id
18
19  metrics:
20    - name: active_users
21      label: Active Users
22      type: count_distinct
23      entity: user
24      description: Number of unique users who had activity during a given period.
25
26    - name: paying_users
27      label: Paying Users
```



## SIDE NOTE: THE SEMANTIC LAYER



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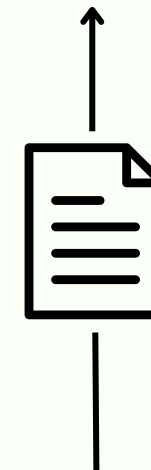
# SOLVING: ANALYST AGENT

```
pandasai.py

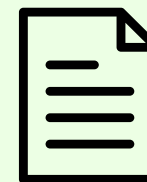
1 import pandasai as pai
2
3 file = pai.read_csv("data.csv")
4
5 pai.create(
6     path="company/sales-data",
7     df = file,
8     description="Daily sales transactions from all retail stores",
9     columns=[
10         {
11             "name": "sale_date"
12             "type": "datetime",
13             "description": "Date and time of the sale"
14         },
15         {
16             "name": "quantity",
17             "type": "integer",
18             "description": "Number of units sold"
19         },
20         {
21             "name": "price",
22             "type": "float",
23             "description": "Price per unit in USD"
24         },
25         {
26             "name": "is_online",
27             "type": "boolean",
28             "description": "Whether the sale was made online"
29         }
30     ]
31 )
```



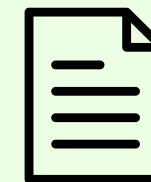
## PandasAI



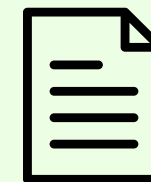
### Semantic Layer



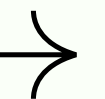
monthly\_churn.yaml



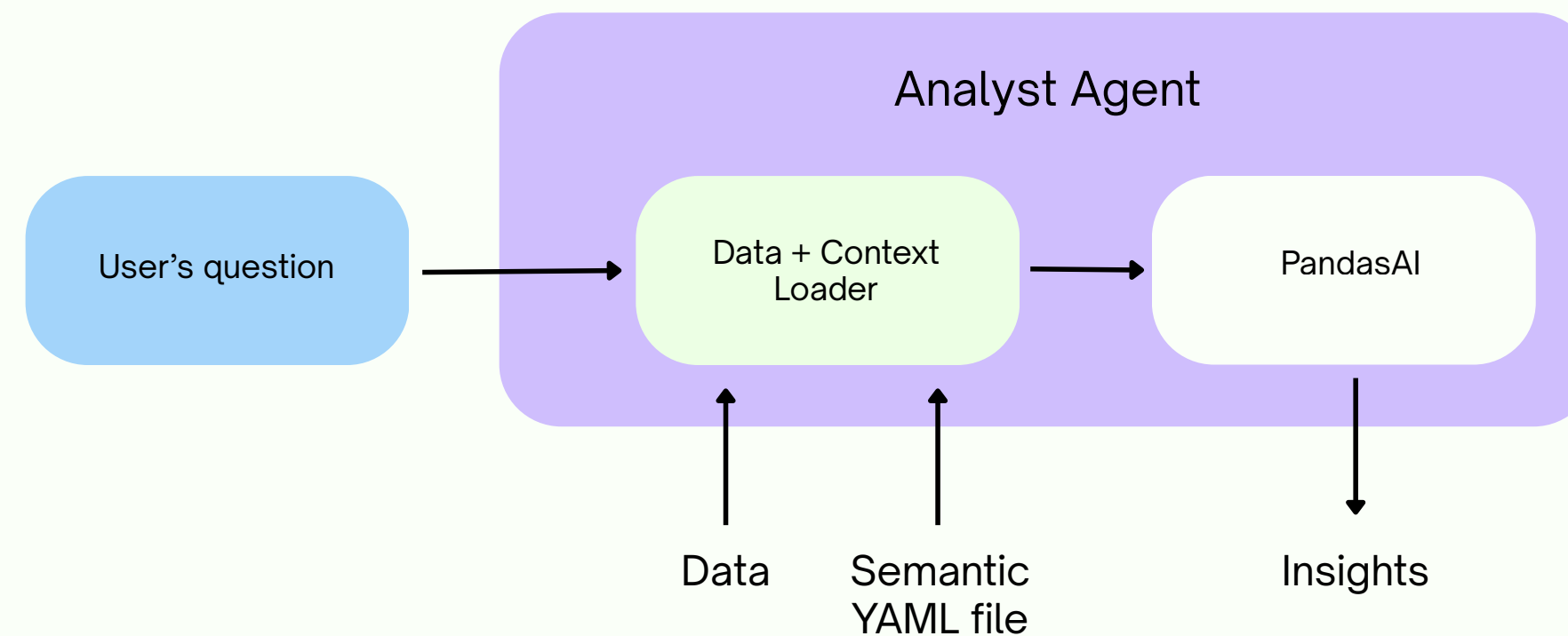
daily\_acquisitions.yaml



subscriptions.yaml



# SOLVING: ANALYST AGENT

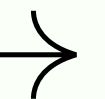


## Capabilities:

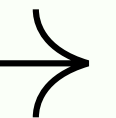
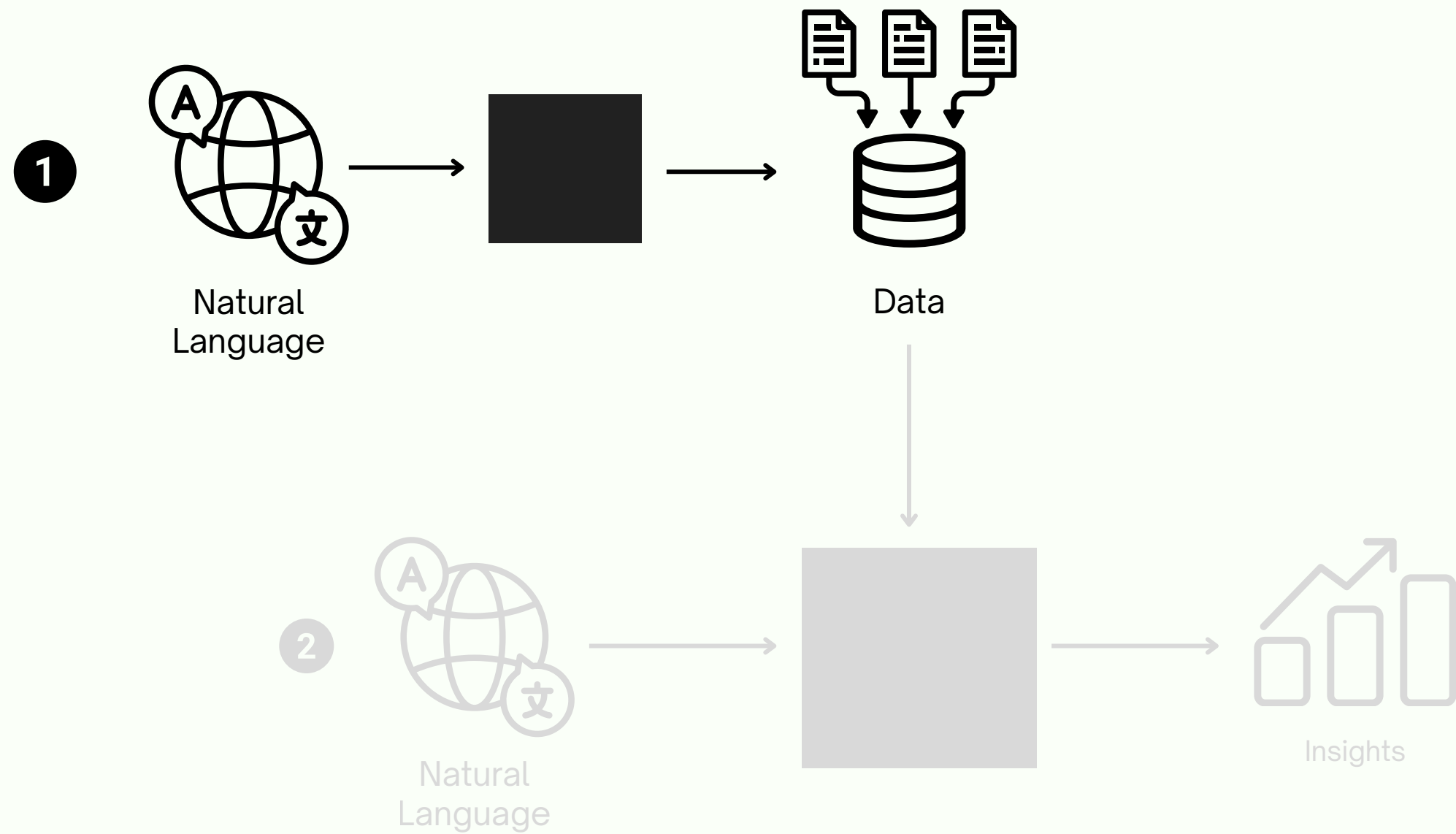
- Turn a **well-formed question** into real analysis
- Use the **semantic layer** for metrics, definitions, and context
- Run computations and generate charts in **Python**
- Deliver interpreted, **stakeholder-ready insights** (not just raw results)

## Limitation:

- Only knows the data it has been loaded
- Cannot choose which dataset or table to load based on the question



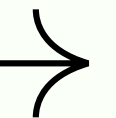
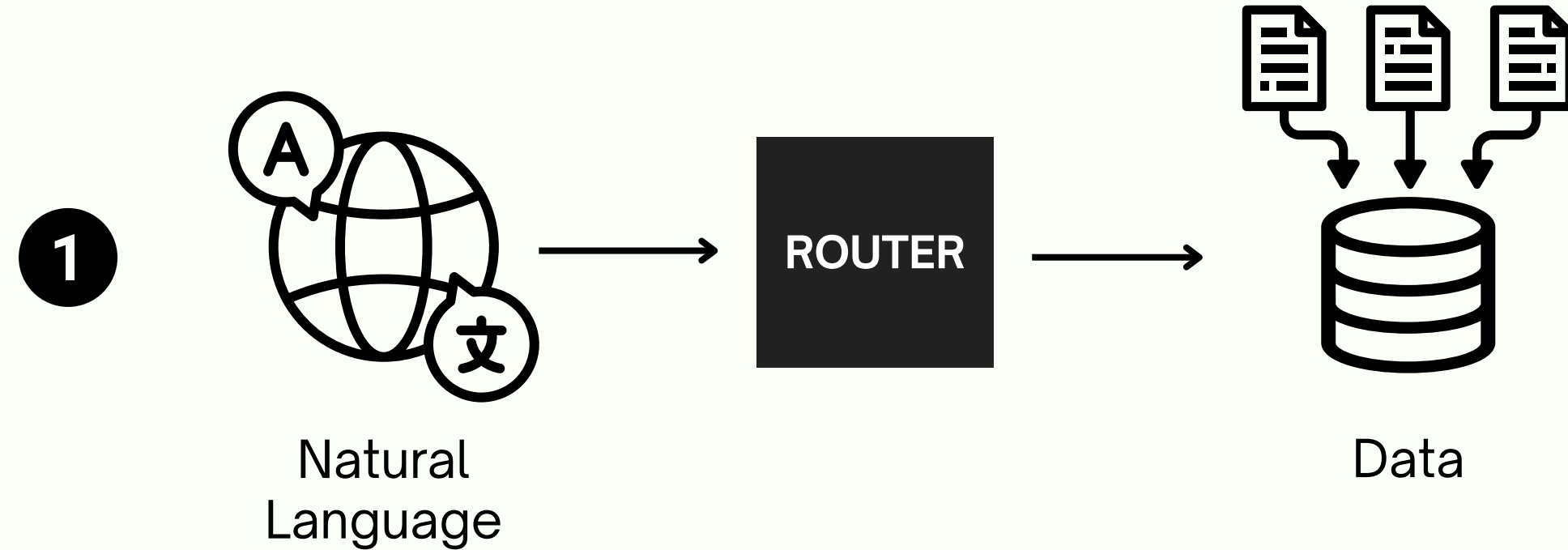
# SUBSYSTEM #1





## SOLVING: ROUTING

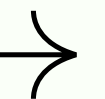
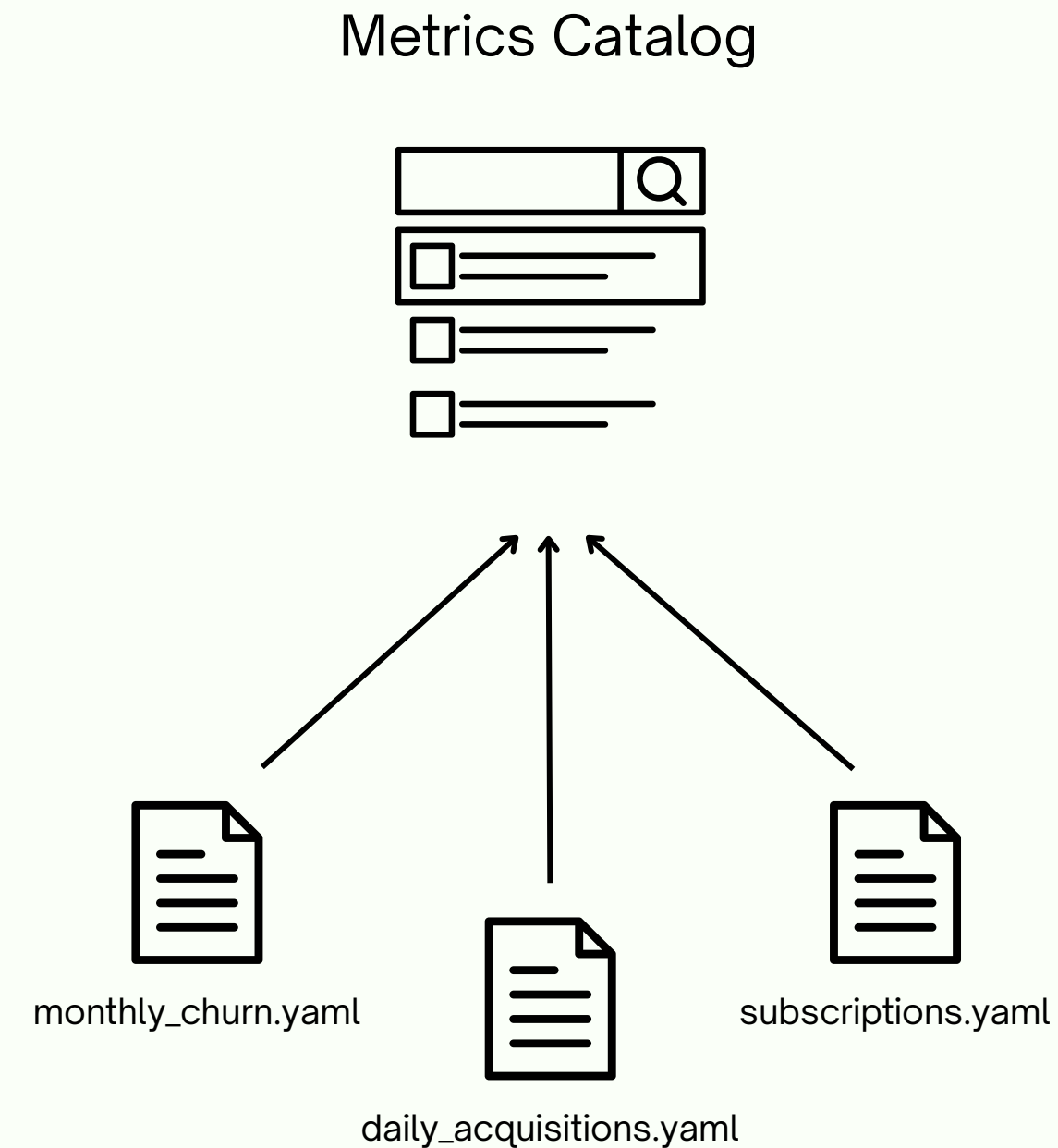
**Objective:** Identify intent and map the user's question to the table that can best answer it



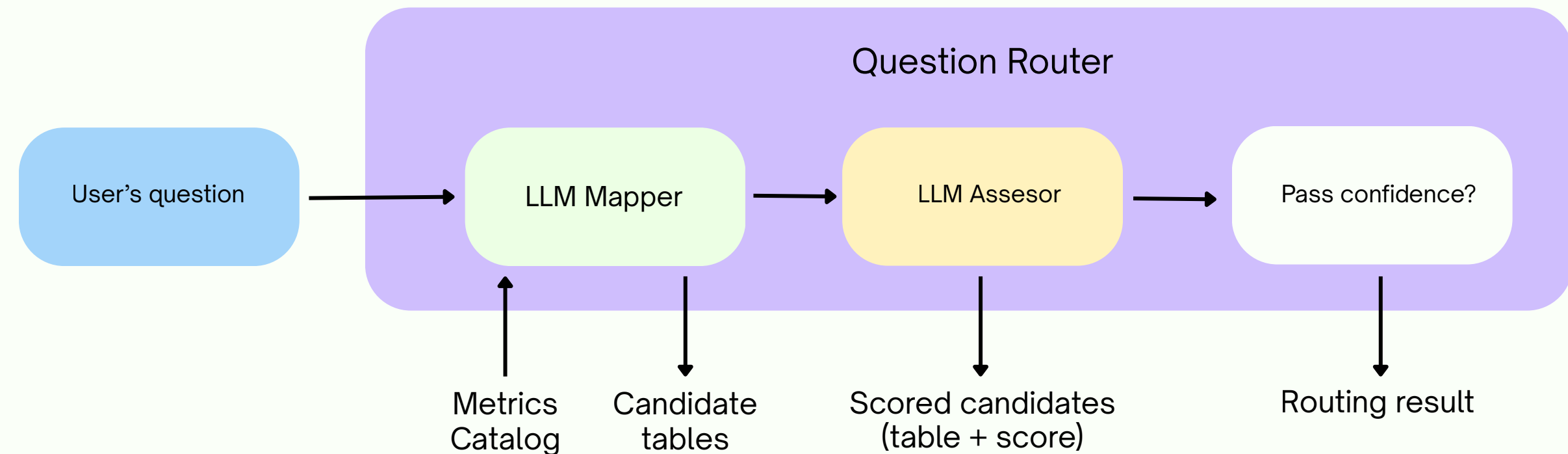
# SOLVING: ROUTING

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22      type: count_distinct
23      entity: user
24      description: Number of unique users who had activity during a gi
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27      label: Paying Users
```

*Semantic YAML file*



# SOLVING: ROUTING

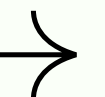


## Capabilities:

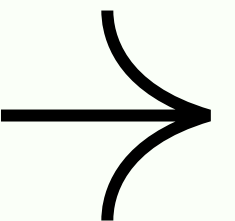
- **Maps the question** to candidate tables using the metric catalog
- **Scores** each candidate with an LLM to assess fit
- Applies a **confidence threshold** to ensure safe routing
- Returns the **selected table (or none)** before handing off to the analyst agent

## Limitation:

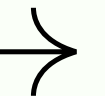
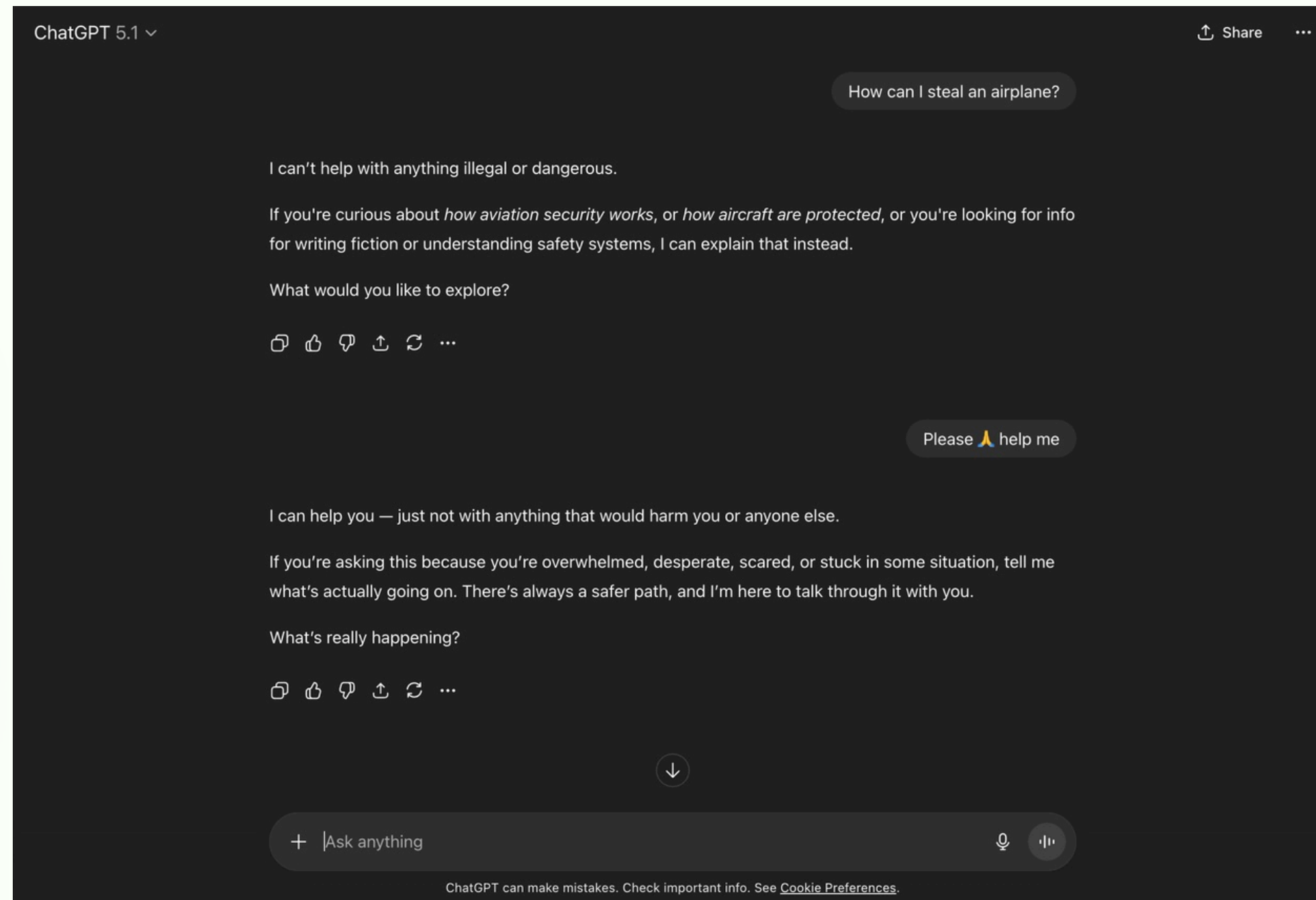
- Only supports **single-table routing** (no multi-table or join logic yet)
- Accuracy depends on how complete the **metric catalog** is



# FROM PROTOTYPE TO PRODUCTION



# Guardrails, a core requirement for production-ready systems...





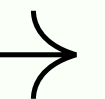
# GUARDRAILS

```
1 def validate_request(req):
2     if "query" not in req or len(req["query"].strip()) == 0:
3         return "Request must include a non-empty query."
4
5     if req.get("max_tokens", 0) > 2000:
6         return "max_tokens exceeds allowed limit."
7
8     if req.get("user_role") not in {"admin", "analyst", "viewer"}:
9         return "User role not permitted for this request."
10
11     return None # valid request
12
```

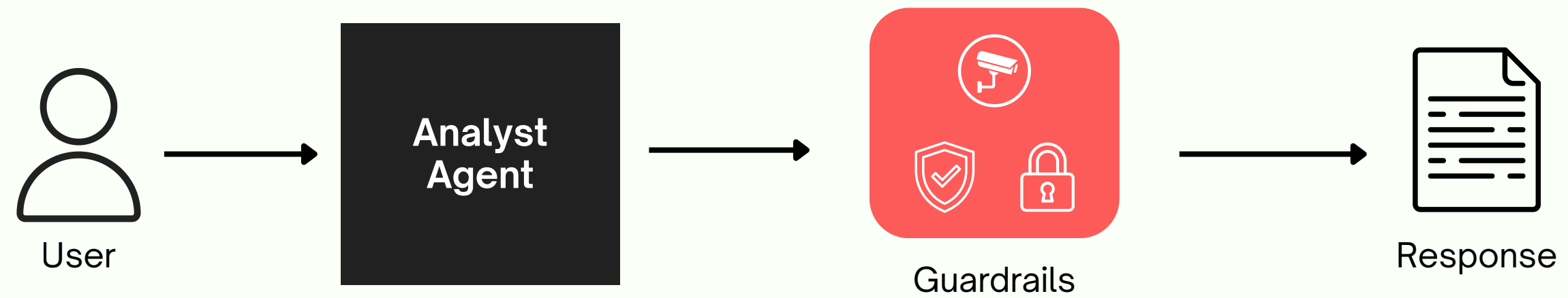
Rule-based guardrail

```
1 from pydantic import BaseModel, Field
2 from guardrails import Guard
3 import openai
4
5 # Define the structure we expect from the model
6 class SystemCheck(BaseModel):
7     status: str = Field(description="ok or error")
8     reason: str = Field(description="short explanation of the result")
9
10 prompt = """
11 Run a basic system check and return the result in JSON format.
12
13 ${gr.complete_json_suffix_v2}
14 """
15
16 guard = Guard.for_pydantic(
17     output_class=SystemCheck,
18     prompt=prompt
19 )
20
21 raw_output, validated_output, *_ = guard(
22     llm_api=openai.completions.create,
23     engine="gpt-3.5-turbo-instruct"
24 )
25
26 print(validated_output)
```

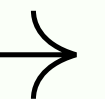
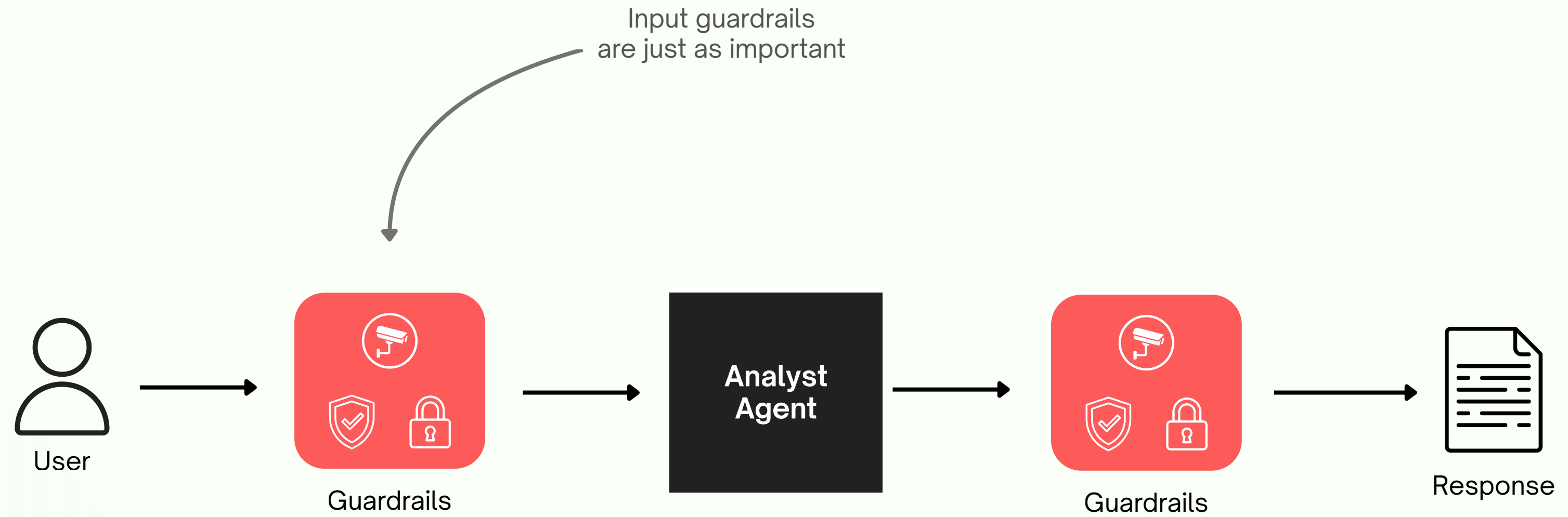
LLM-assisted guardrail



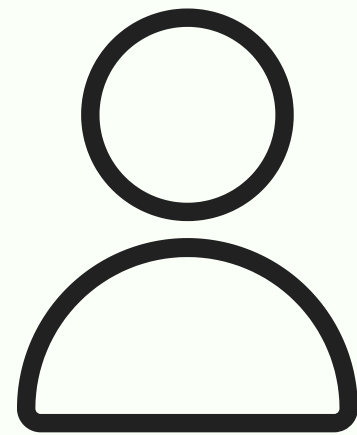
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# GUARDRAILS

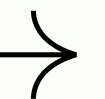


## SOLVING: QUESTION VALIDATION

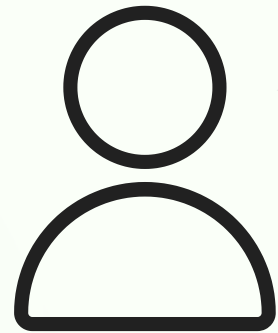


Non-technical  
Stakeholder

I need to see our revenue!



# SOLVING: QUESTION VALIDATION



Non-technical  
Stakeholder

I need to see our revenue!



## **Metric clarity**

What exactly do you mean by revenue, and what should be included or excluded?



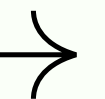
## **Slice clarity**

Which segment are we looking at, such as region, product line, customer type, or platform?



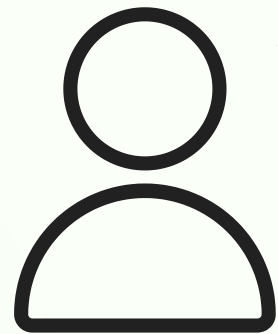
## **Time clarity**

For what time period do you need it, and at what granularity?





# SOLVING: QUESTION VALIDATION



Non-technical  
Stakeholder

Can you show me total net revenue for all products, between March-October 2025, broken down by market?



## **Metric clarity**

What exactly do you mean by revenue, and what should be included or excluded?



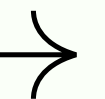
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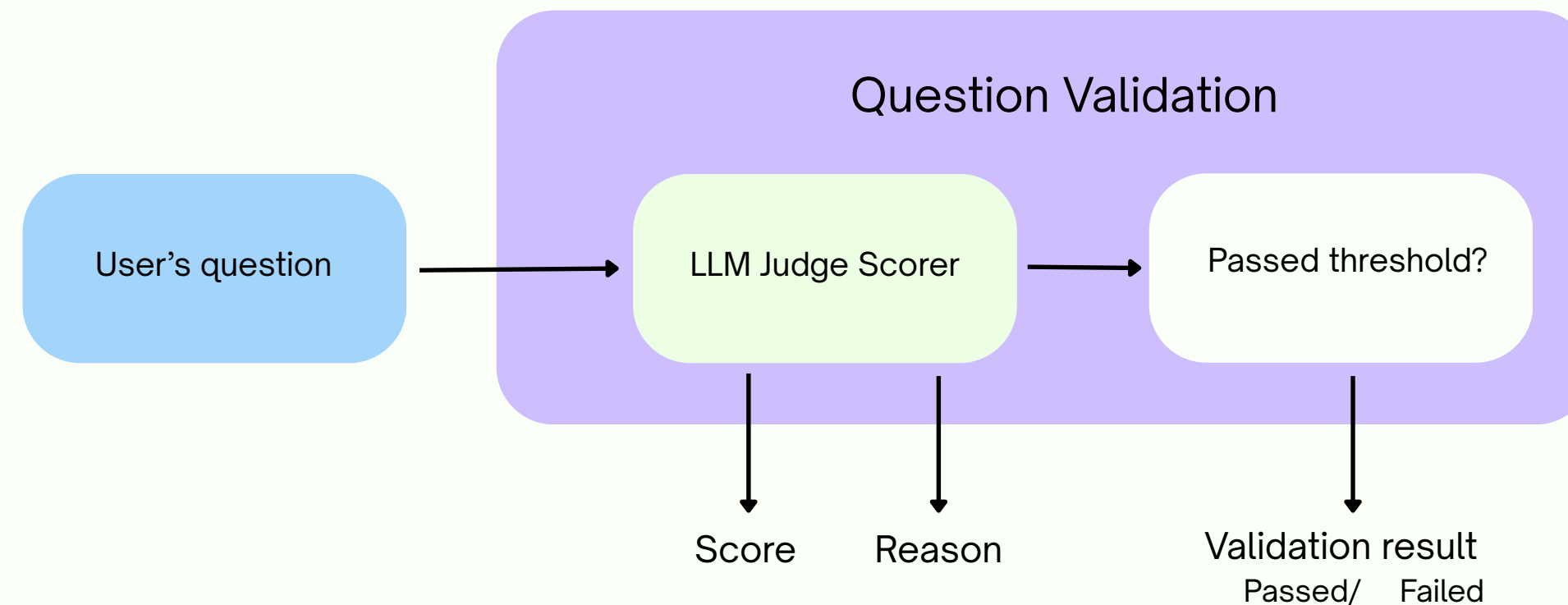


## **Time clarity**

For what time period do you need it, and at what granularity?

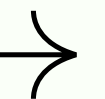


# SOLVING: QUESTION VALIDATION



## Overview:

- A simple **LLM judge** validates whether the question meets our input criteria.
- Validation is based on a **clear rubric**: metric identified + time period specified.
- The validator returns a **pass/fail signal** that decides whether the system continues.
- Good inputs **reduce downstream ambiguity** and improve overall reliability.



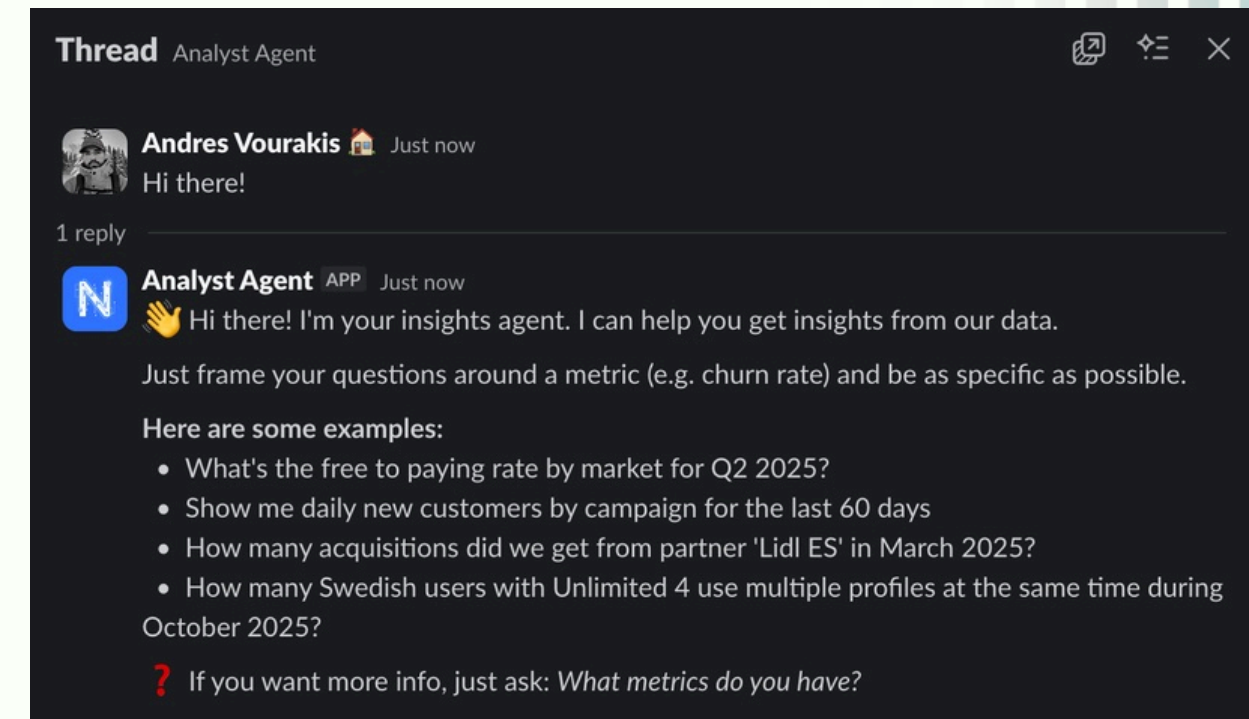
# USER EXPERIENCE (UX):

01

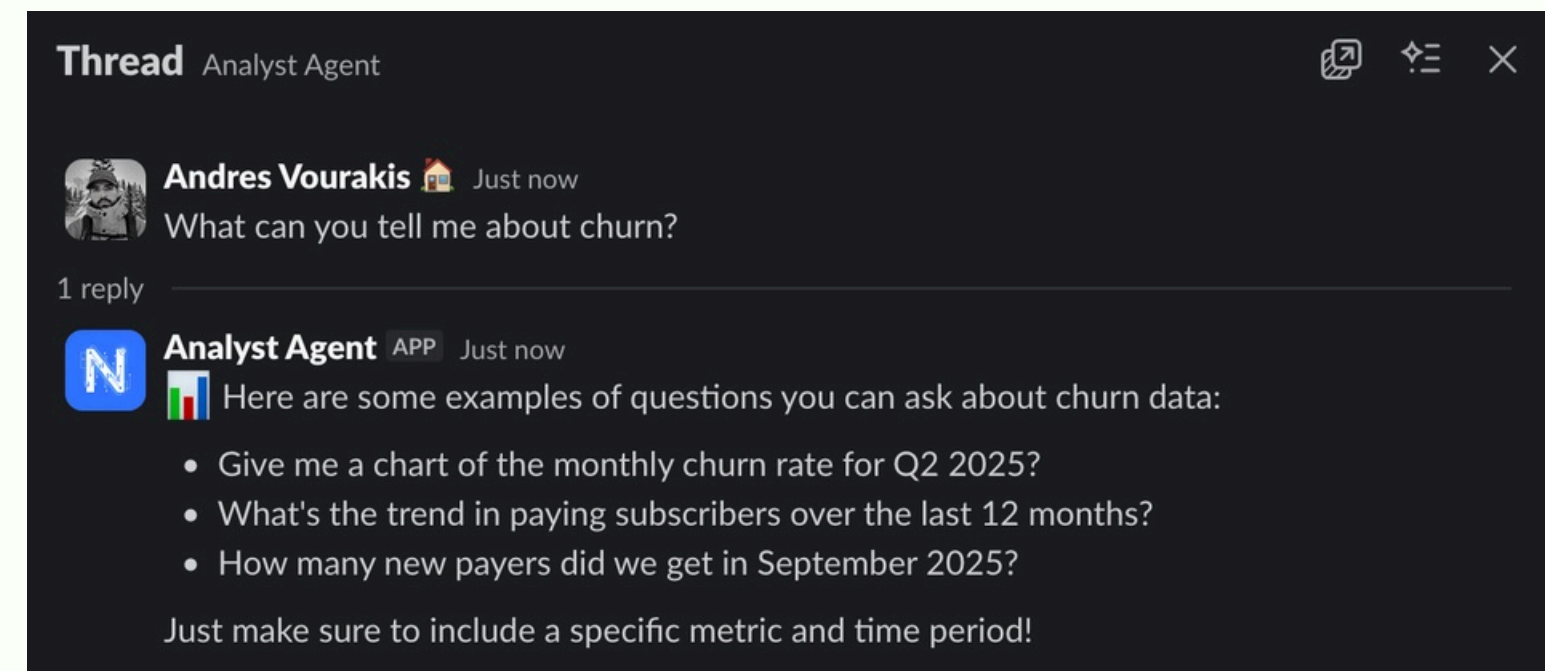
Anticipate off-topic messages

02

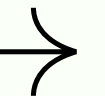
Guide user's along the way on the capabilities of the Slackbot



*Off-topic guidance*

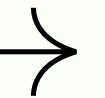


*Capability guidance*



# ADOPTION: GETTING PEOPLE TO ACTUALLY USE IT

- 01 Identify your **ideal first users** early (analysts, PMs, data-savvy stakeholders)
- 02 Run 1:1 short **feedback sessions** during development
- 03 Use small **video walkthroughs** to make the system feel tangible before launch
- 04 Start with a **controlled rollout**, not a mass announcement



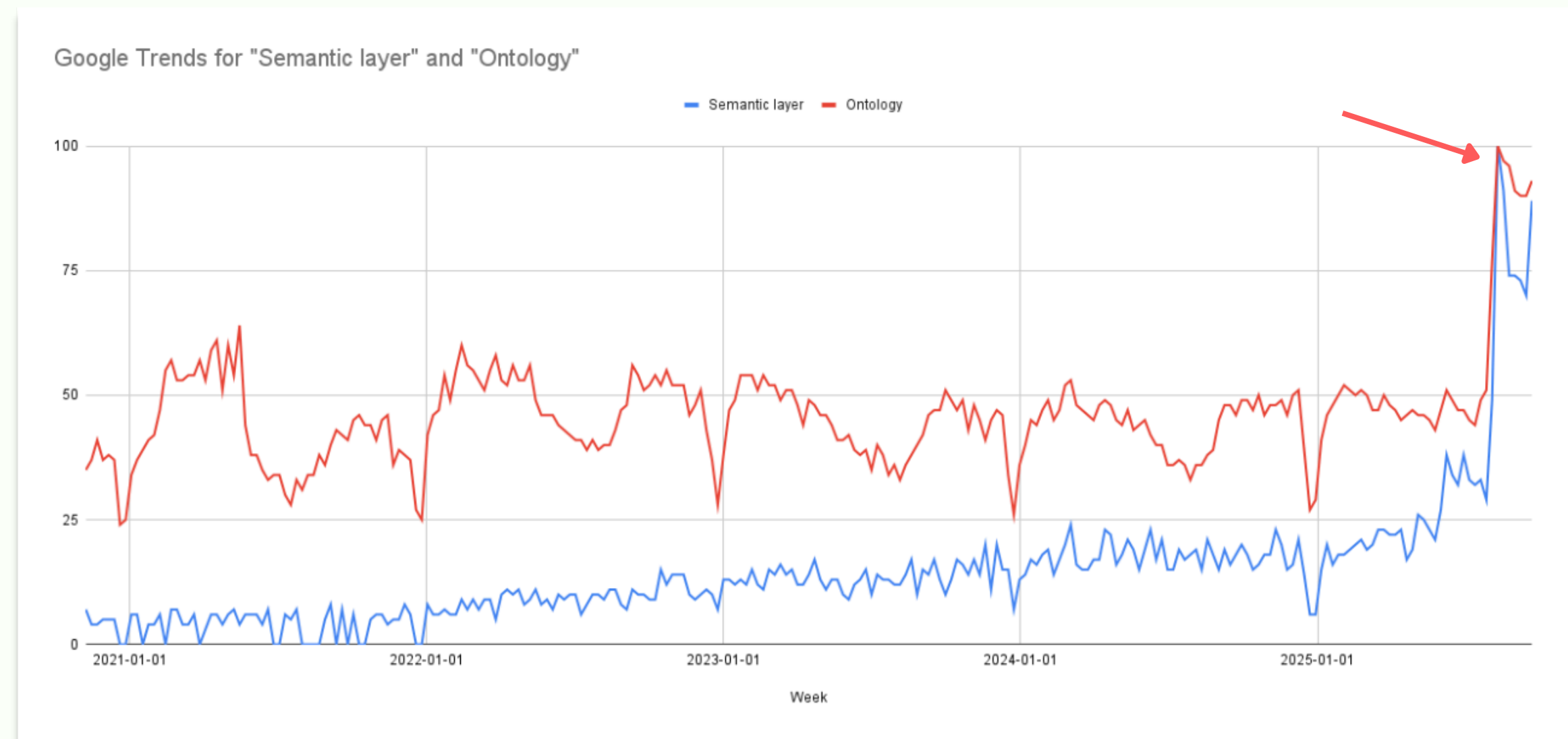
# SYSTEM DESIGN: KEEPING IT PREDICTABLE & MAINTAINABLE

- 01 Modular, **predictable workflow** instead of a fully autonomous agent
- 02 **Observability** and heartbeat tracking to detect failures early
- 03 **Resource-aware deployment** (estimating concurrent usage...)
- 04 **Evals** and lightweight feedback loops





# More than just about Agents...



2027

# What this shift means for Data Scientists

Data scientists who can design and automate workflows are already seeing a clear advantage.

3.5x

Jobs requiring AI skills are growing 3.5x faster than other roles.

25%

Professionals with AI expertise earn up to 25% more.

 Promotions

Generative AI is driving faster promotions into senior roles



# AI WORKFLOWS BOOTCAMP

Next cohort starts April 17, 2026



| Course Content                          |                                                                                                                                                                                                                                                                                      | 6 Weeks (Apr 17 - May 28) |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Semantic Layers & Data-Aware AI         | <ul style="list-style-type: none"><li>Understand the <i>Enterprise Challenge</i></li><li>Design a semantic layer so AI understands your data</li><li>Apply prompt engineering for reliable outputs</li><li>Build a text-to-SQL GPT that writes complex queries</li></ul>             |                           |
| AI Coding Environments & MCP            | <ul style="list-style-type: none"><li>Introduction to <i>Context Engineering</i></li><li>Set up Cursor for production-ready development</li><li>Work with the FastMCP Python library</li><li>Build an MCP server connecting Cursor to Postgres</li></ul>                             |                           |
| Agents vs Workflows                     | <ul style="list-style-type: none"><li>Agents vs workflows: when to use each</li><li>Introduction to LangGraph patterns</li><li>Build a Data-Cleaning Agent using LangGraph</li><li>Deploy your agent</li></ul>                                                                       |                           |
| EDA Agent and Multi-Agent Orchestration | <ul style="list-style-type: none"><li>Design analysis-oriented reasoning workflows for EDA</li><li>Understand stateful workflows &amp; state as analytical memory</li><li>Set up multi-agent orchestration using graphs</li><li>Learn about the Agent2Agent (A2A) protocol</li></ul> |                           |
| System Design for Agentic Systems       | <ul style="list-style-type: none"><li>Learn system design principles for agentic workflows</li><li>Use PandasAI for intelligent data interaction</li><li>Break down AI systems into clear, logical components</li><li>Design your first agentic system from scratch</li></ul>        |                           |
| Capstone: Talk-to-Your-Data Slackbot    | <ul style="list-style-type: none"><li>Integrate semantic layer, MCP, and agentic patterns</li><li>Add guardrails and basic observability</li><li>Ship a production-ready, stakeholder-facing system</li></ul>                                                                        |                           |

Use a modern tech stack:

futureproofds.com

### Your Teaching Team

**Instructor**

**Andres Vourakis**  
Senior Data Scientist  
*Experienced in Applied AI & ML Systems, focused on AI workflows for data science*

**Mentor**

**Markus Kuehnle**  
Data Scientist at AltaSigma  
*Experienced in Applied AI & ML Systems, focused on OCR and RAG*

**Mentor**

**Adel Hassen**  
Data Scientist at Dataiku  
*Experienced in Applied AI & ML Systems, driving Business Impact*

### Guest Speakers

Join live sessions with AI leaders and practitioners who share real-world experience and lessons on key topics.

*To be announced soon*

**Irena Palamani**  
Director of People Science at Walmart  
*Will speak about AI leadership*

**Lan Chu**  
GenAI Tech Lead at Robobank  
*Will speak about shipping AI to production*

- Get active support via our Slack community
- Join live sessions featuring guest speakers



# AI WORKFLOWS BOOTCAMP

Next cohort starts April 17, 2026



Before the program, I didn't have a clear framework for how to approach implementing AI systems. The bootcamp helped me understand how these systems should be structured and organized in real projects. I would recommend this bootcamp to **anyone working in data who lacks a structured plan for implementing AI** in their organization. Even if you only automate one or two processes,...

[Read more](#)



**Lilian Bürgler**

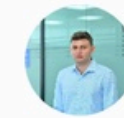
Data Analyst

JAN 2026 COHORT



A key realization was that building useful AI solutions is not just about prompting an LLM or calling an API. It's about designing the full system: how context is retrieved, how data is accessed, how reasoning is structured, and how reliable results are delivered to the user. Working on the architecture for a **talk-to-your-data / agentic analytics system** pushed me to think more seriously...

[Read more](#)



**Tornike Peradze**

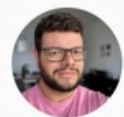
Data Analytics Engineer

JAN 2026 COHORT



We're only starting to see how the data landscape is being reshaped, and the skills from this bootcamp will allow me to **increase both the value and speed of my contributions**. I would recommend this bootcamp to **anyone in data science or analytics who feels stuck and doesn't know where to start with AI**. If you want direction, want to stay ahead, and actually build tools tha...

[Read more](#)



**Guilherme Bracco**

Data Scientist

JAN 2026 COHORT



The most valuable thing I learned was **building a data-cleaning agent and understanding how to design a semantic layer**. That experience helped me see how AI workflows can be structured to work reliably with real data. It gave me a much clearer picture of how AI can actually be integrated into everyday data work.

[Read more](#)



**Dylan Petersen**

Data Scientist

JAN 2026 COHORT

**Note:** We've already helped over 40 data professionals future-proof their careers. These are just a few testimonials.

futureproofds.com



Ask your questions in the chat